SOHINI DUTTA

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PROFILE

Final-year PhD researcher in astrophysics (expected submission: 31 March 2027, expected thesis defence: July 2027) specializing in Bayesian inference, probabilistic modeling, and uncertainty quantification for high-dimensional, systematics-dominated data. Builds statistical and machine-learning pipelines (MCMC, Gibbs sampling, ANN emulators, time-series forecasting) to extract robust, calibrated estimates from noisy real-world data where no clean reference signal exists. Prior applied experience as a data scientist in marketing decisions at American Express. Currently developing applied fluency in LLM-based and agentic reasoning systems to extend this background into large-scale forecasting and estimation problems.

KEY TECHNICAL SKILLS

- **Probabilistic modeling & uncertainty quantification**: Bayesian inference, MCMC, Gibbs sampling, calibration of systematic effects in noisy, high-dimensional data
- **Forecasting & temporal data**: ARIMA time-series modeling; parameter estimation from indirect, evolving observational data with no stable ground truth
- **Machine learning**: ANN-based emulators, gradient-boosted classifiers for industry risk modeling, CNNs for image-based classification
- **Programming & pipelines**: Python, C/C++, SQL, PySpark, Git/GitHub, Linux, large-scale simulation pipelines
- **Currently building**: applied familiarity with LLM tool-use and agentic workflows

RESEARCH EXPERIENCE

PhD Researcher, Bayesian Cosmology | Jodrell Bank Centre for Astrophysics, University of Manchester *Oct 2023 – Present*

- Developed Bayesian power-spectrum estimation methods using Gaussian-constrained realizations and Gibbs sampling to model and marginalize instrumental systematics in the HERA radio interferometer's Hydra pipeline. This is directly analogous to calibrating estimates under structured, non-random noise.

Master's Student, ML-based Cosmological Parameter Estimation | Indian Institute of Technology, Indore *Aug 2021 – 2023*

- Built ANN-based emulators combined with Bayesian inference to estimate cosmological reionization parameters from noisy multi-wavelength survey data.
- Applied ARIMA time-series modeling to forecast structure in observational data — early, direct experience with the temporal-forecasting problem class.
- Co-authored work developing Bayesian neural network (BNN) emulators that produce calibrated posterior uncertainty, rather than point-value predictions, for 21-cm signal inference, directly addressing the same uncertainty-propagation problem this role targets.

Research Intern, Pulsar Timing | National Centre for Radio Astronomy, Pune *May – Aug 2021*

- Searched for intermittent signal behaviour ('nulling') in millisecond pulsars from live radio-telescope data.

INDUSTRY EXPERIENCE

Analyst, Data Science — Marketing Decisions, Global Decision Science | American Express, Gurugram, India *Oct 2022 – Jun 2023*

- Built and evaluated gradient-boosted classifiers for marketing decision making on live production data, working within a regulated, high-stakes decision-making environment. This is the same kind of accountable, real-world deployment context this role targets.

SELECTED PUBLICATIONS

- Dutta, Bull, Burba, Wilensky, Zhang, Nasirudin — "Bayesian power spectrum estimation with modelling of systematic effects in delay-fringe rate space."
- Bull, El-Makadema, Garsden, et al. (incl. Dutta) — "RHINO: A large horn antenna for detecting the 21cm global signal."
- Dutta, Majumdar, Tiwari, Murmu — "Interpreting multi-wavelength observations of the Epoch of Reionization from next-generation telescopes," URSI-RCRS 2022.
- Mahida, Yadav, Majumdar, Noble, Murmu, Dasgupta, Dutta, Tiwari, Shaw — "From ANN to BNN: Inferring reionization parameters using uncertainty-aware emulators of 21-cm summaries," JCAP 2025 (co-author).

Full list of publications and invited talks available at sohinidutta.com and on request.

EDUCATION

PhD in Astronomy | University of Manchester, UK (Supervisor: Dr. Phil Bull) *Oct 2023 – expected July 2027*

MSc in Astronomy | Indian Institute of Technology, Indore, India — CGPA 9.82/10 *Aug 2020 – Aug 2022*

BSc (Hons) in Physics | University of Delhi, India — CGPA 7.24/10 *2016 – 2019*

OTHER

- Graduate Teaching Assistant, Object-Oriented Programming in C++, University of Manchester (2024)
- National Rank 344/~13,000, Joint Admission Test for MSc (IIT JAM), 2020
- Offered PhD positions at Cambridge, Manchester, Heidelberg, and Swinburne